LESSON 2.3

Describe the Data Using the Median

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find the median of a data set and use it to describe what is typical.

WHY THIS MATTERS You're the stats analyst for the school basketball team. Coach needs a report on the team's scoring: in the last 7 games they scored 42, 38, 55, 41, 38, 44, and 40 points. Which average best represents the team's typical performance?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Median *Spanish: Mediana*	The middle number when you put them in order.	2, 6, 9 . . . median is 6
Mode *Spanish: Moda*	The number that shows up most often.	Data: 2, 3, 3, 5, 7 → 3 appears twice (most often), so mode = 3
Outlier *Spanish: Valor atípico*	A number that is much bigger or smaller than the rest.	2, 3, 4, 40 → 40
Data distribution *Spanish: Distribución de datos*	How the data looks: where it sits and how spread out it is.	If most values cluster in the center with fewer at the ends, the distribution is symmetric
Variability *Spanish: Variabilidad*	How spread out the numbers are.	8, 9, 10, 11 has low variability (close together); 2, 10, 25, 40 has high variability (far apart)
Measure of center *Spanish: Medida de tendencia central*	One number that describes a typical value in a data set.	The median is one of them

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Find the median of 7, 2, 9, 4, 6.

1. Order: 2, 4, 6, 7, 9.
2. Cross off 2 and 9, then 4 and 7.
3. One value is left: 6.

Answer: Median = 6

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

New data set, out of order: 13, 3, 7, 10, 7. Put it in order first.

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- In order, the values are _____ .
- There are _____ values, so the middle one is in position _____ .
- The median is _____ , which means _____ .
- About half the values are below _____ .

Word bank median · order · least · greatest · middle · half · typical · center · position · count · outlier · data

WATCH OUT!

A common mistake

A common mistake in Mean, Median, and Mode is finding the median without ordering the data first

LESSON 2.3

Describe the Data Using the Median

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

Put the steps for finding median in the correct order.

Groups: **Steps to find the median (in order)**

_____ Step 1: Order the data from least to greatest

_____ Step 2: Count the total number of values

_____ Step 3: Find the middle value (or average of two middle values)

_____ Step 4: The middle value is the median

HINT You must order the data FIRST before finding the middle.

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

Find the mean of: 10, 14, 8, 12, 16

(A) 10
(B) 12
(C) 14
(D) 60

HINT Mean means the average of all five numbers — you'll need both a total and how many numbers there are.

Put the data in order _____

WORK

1. **Order** *Smallest to largest.*
2. **Count** *How many values?*
3. **Locate** *Middle, or add and divide.*
4. **Answer** *Say which measure it is.*

3 **Circle the letter of the best answer. Show how you know.**

Find the median of: 3, 7, 2, 9, 5

(A) 3
(B) 5
(C) 5.2
(D) 7

HINT Median = the middle number, but only AFTER the list is in order. Don't pick the middle of the unsorted list.

Put the data in order _____

WORK

1. **Order** *Smallest to largest.*
2. **Count** *How many values?*
3. **Locate** *Middle, or add and divide.*
4. **Answer** *Say which measure it is.*

4 **Circle the letter of the best answer. Show how you know.**

Find the mode of: 4, 7, 4, 9, 2, 4, 8

(A) 2
(B) 4
(C) 7
(D) 9

HINT Mode is about repeats — which value shows up in the list more times than any other?

Put the data in order _____

WORK

1. **Order** *Smallest to largest.*
2. **Count** *How many values?*
3. **Locate** *Middle, or add and divide.*
4. **Answer** *Say which measure it is.*

5 **Circle the letter of the best answer. Show how you know.**

Find the mean of: 6, 8, 10, 12, 14

(A) 8
(B) 10
(C) 12
(D) 50

HINT Watch out: one tempting answer is just the total of the list. The mean asks you to go one step further.

Put the data in order _____

WORK

1. **Order** *Smallest to largest.*
2. **Count** *How many values?*
3. **Locate** *Middle, or add and divide.*
4. **Answer** *Say which measure it is.*

PART 3 Explain and correct

reasoning

6 **Find the mistake. Explain it, then show the correct work.**

The Mean Mistake (forgot to divide)

1. **Data set** A player scored 6, 8, 10, and 6 points in 4 games.
2. **Goal** Find the MEAN number of points.
3. **Add the values** 6 + 8 + 10 + 6 = 30
4. **Answer** Mean = 30

Which step has the mistake? Explain what went wrong, then show the correct work.

HINT Mean = add then DIVIDE. The total of 30 is just the sum, not the mean.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

The mean (43) is higher than the median (41). Why? Which better represents a typical game?

SENTENCE STARTER The mean is higher than the median because the high score of ___ pulls the mean ___. The ___ better represents a typical game because ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 2.3

Describe the Data Using the Median

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Find the mean, median, and mode of each data set. Round each mean to the nearest hundredth when needed.

Data Set	Mean	Median	Mode
5, 8, 8, 10, 14			
20, 25, 25, 30, 25, 35			
100, 110, 90, 100, 115, 95			

SHOW YOUR WORK

2 **Write the letter of the matching item on each line.**

Match each measure to its definition.

_____ Mean

_____ Median

_____ Mode

_____ Range

(A) Sum ÷ Count

(B) Middle value

(C) Most frequent

(D) Max − Min

3 **Mark and label each value on the number line.**

Data: 20, 25, 30, 35, 40. Place the mean and median on the number line.



SHOW YOUR THINKING

PART 2 Apply it

real-world problems

4 **Circle the letter of the best answer. Show how you know.**

Data set: 4, 6, 6, 8, 10, 12. What is the median?

(A) 6
(B) 7
(C) 7.67
(D) 8

Put the data in order _____

WORK

PART 3 Explain and correct

reasoning

5 **Find the mistake. Explain it, then show the correct work.**

The Even-Count Median Mistake

1. **Data set** Points per game, in order: 3, 8, 10, 14, 16, 20 (6 values)
2. **Goal** Find the MEDIAN number of points.
3. **Pick the middle** With 6 values the student grabs a single middle number and says median = 10.
4. **Answer** Median = 10

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

The mean (43) is higher than the median (41). Why? Which better represents a typical game?

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LESSON 2.3

Describe the Data Using the Median

Challenge · Extension

MASTERY CHECK
- ☐ Exceeds
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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Complete the table.

Data Set	Mean	Median	Which is better & why?
10, 12, 11, 13, 12			
10, 12, 11, 13, 50			

SHOW YOUR WORK

PART 2 Explain and correct

reasoning

2 Find the mistake. Explain it, then show the correct work.

Find the Stats Error

1. **Data set** 12, 15, 18, 15, 20
2. **Find mean** Sum = 80, Count = 5, Mean = 80 ÷ 5 = 16
3. **Find median** Ordered: 12, 15, 15, 18, 20 → Median = 15
4. **Find mode** Mode = 18 (the biggest number)

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

3 Answer in complete sentences.

A basketball player scored 15, 18, 16, 42, and 14 points in 5 games. Find the mean and median. Which better represents the player's typical game? Explain how the outlier affects each measure.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

The mean (43) is higher than the median (41). Why? Which better represents a typical game?

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How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help